

Locked and loading megathrust linked to active subduction beneath the Indo-Burman Ranges

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Figures with station names

For clarity, the 4-letter station names were omitted from the figures in the paper. They are included here with all of the station names.

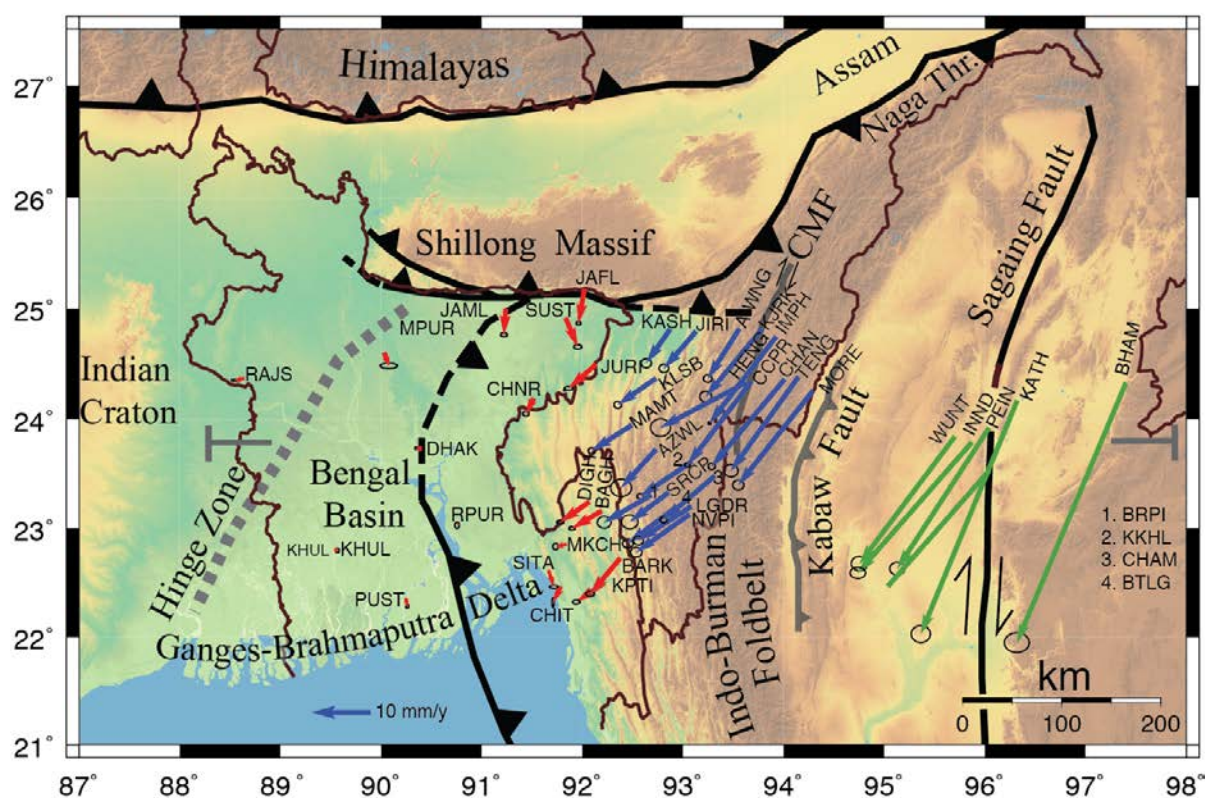


Figure S1. Topographic map of the Ganges-Brahmaputra Delta and Indo-Burman Foldbelt showing velocity arrows for GPS stations in Indian frame of reference. Red, blue and green arrows are new stations from Bangladesh, stations from India¹¹, and stations from Myanmar⁹, respectively. Same as Figure 1 with the addition of 4-letter station names. Numbers 1-4 at base of arrows indicate station listed on right.

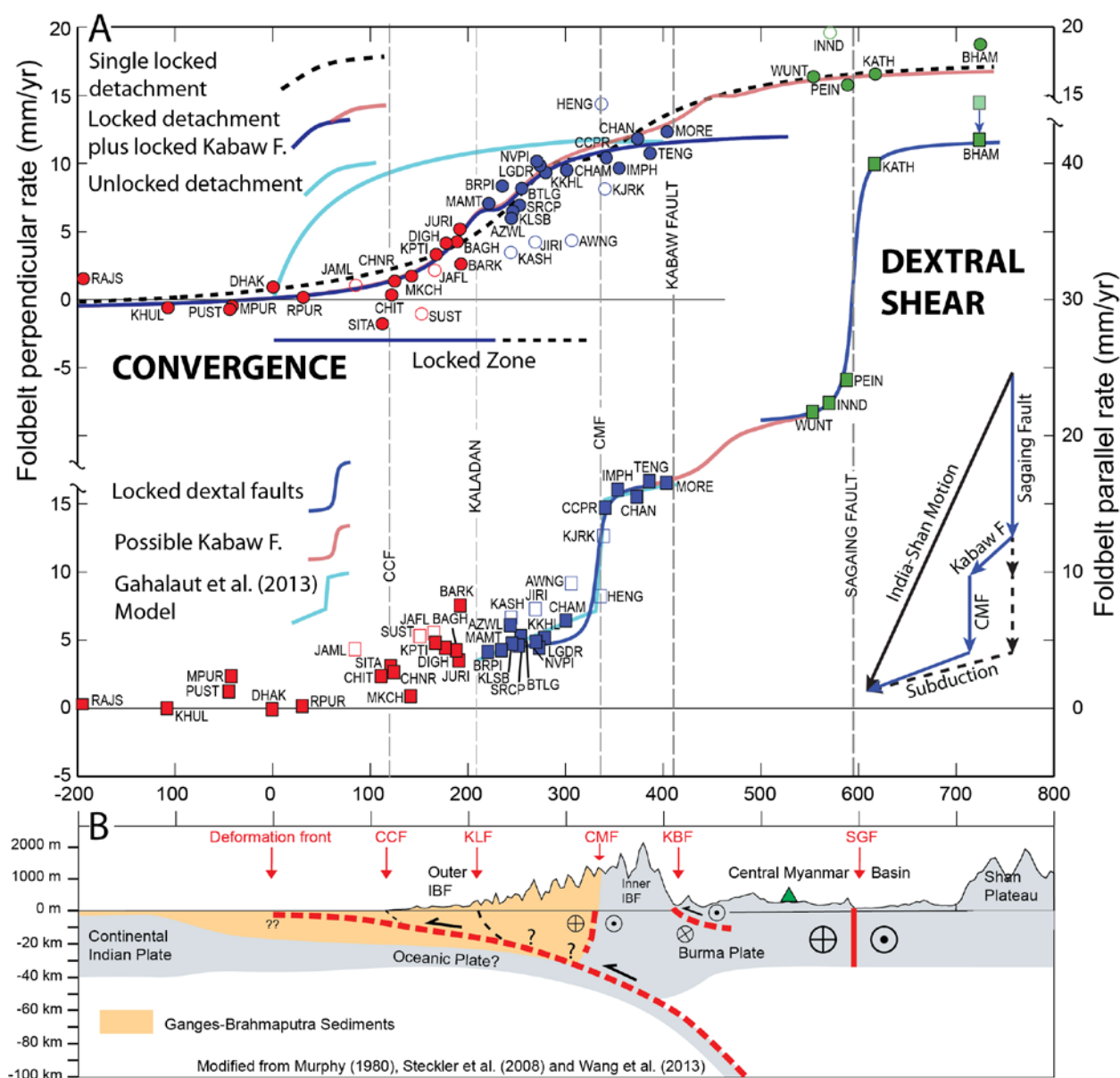


Figure S2. Foldbelt perpendicular convergence and parallel dextral shear components of velocity field. All elements are the same as in Figure 2, but the individual 4-character GPS site names have been added.

Table S1. GPS stations velocities in ITRF2008 reference frame³³ after adjustment by Helmert transformation to merge data sets. Continuous Bangladesh and campaign Myanmar stations were processed together from RINEX files. Continuous and campaign velocities for Indian station were taken from Gahalaut et al.¹⁰.

	ITRF2008						Velocity wrt India	
	Longitude	Latitude	N	E	N σ	E σ	N	E
	(°E)	(°N)	(mm/y)	(mm/y)	(mm/y)	(mm/y)	(mm/y)	(mm/y)
Bangladesh Stations (continuous measurements from 2003 or 2007 to 2013)								
DHAK	90.401	23.727	35.83	39.86	0.08	0.10	0.11	-0.92
CHIT	91.790	22.465	33.00	40.10	0.33	0.08	-2.73	-1.42
KHUL	89.535	22.803	35.56	41.42	0.15	0.12	-0.14	0.55
RAJS	88.638	24.371	35.31	38.58	0.07	0.28	-0.36	-1.54
SUST	91.831	24.919	30.75	42.83	0.15	0.28	-4.98	2.08
PUST	90.244	22.357	34.32	41.47	0.11	0.11	-1.40	0.28
JAFL	92.019	25.179	29.85	39.70	0.12	0.17	-5.88	-1.02
JAML	91.237	25.000	31.29	40.26	0.14	0.24	-4.44	-0.31
JURI	92.135	24.498	31.60	36.28	0.13	0.32	-4.13	-4.69
CHNR	91.515	24.190	30.04	39.57	0.17	0.22	-2.69	-1.34
MPUR	90.030	24.604	33.44	41.15	0.19	0.60	-2.27	0.76
SITA	91.662	22.615	32.91	42.35	0.11	0.31	-2.82	0.91
KPTI	92.217	22.499	32.43	36.79	0.13	0.26	-3.30	-4.82
BARK	92.375	22.726	29.63	36.41	0.17	0.31	-6.10	-5.17
BAGH	92.192	23.162	32.69	36.30	0.12	0.22	-3.04	-5.10
MKCH	91.845	22.851	35.39	39.48	0.22	0.20	-0.34	-1.93
RPUR	90.763	23.036	35.56	40.92	0.22	0.16	-0.16	-0.18
DIGH	92.069	23.242	32.27	36.36	0.13	0.25	-3.46	-4.99
Myanmar Stations (campaign measurements from 2005 and 2008)								
KATH	96.32	24.17	-4.34	25.61	0.60	0.67	-39.97	-16.53
INND	95.87	23.79	13.26	22.57	0.39	0.54	-22.39	-19.56
PEIN	96.04	23.92	11.55	26.41	0.45	0.53	-24.09	-15.73
BHAM	97.40	24.33	-8.88	23.67	0.69	0.87	-44.45	-18.70
WUNT	95.70	23.84	13.92	25.73	0.51	0.58	-21.74	-16.34
India Stations (velocities from Gahalaut et al., 2013)								
MORE	94.29	24.258	19.55	28.80	0.34	0.40	-16.15	-12.80
TENG	94.146	24.384	19.64	29.96	0.43	0.49	-16.07	-11.56
CHAN	94.009	24.363	20.79	28.98	0.22	0.25	-14.92	-12.51
IMPH	93.925	24.749	20.92	29.93	0.09	0.10	-14.79	-11.43
KJRK	93.785	24.817	24.16	31.62	0.37	0.41	-11.55	-9.68
CCPR	93.679	24.344	21.40	30.45	0.16	0.19	-14.32	-10.96
HENG	93.624	24.331	28.02	26.76	0.62	0.70	-7.70	-14.64
AWNG	93.559	24.821	27.16	35.79	0.26	0.30	-8.56	-5.44

CHAM	93.313	23.480	28.25	32.86	0.22	0.25	-7.47	-8.72
JIRI	93.128	24.796	29.37	35.62	0.30	0.34	-6.35	-5.51
KKHL	93.087	23.611	29.80	32.65	0.19	0.23	-5.93	-8.84
LGDR	93.082	23.202	29.87	32.51	0.30	0.36	-5.86	-9.10
NVPI	93.064	23.129	29.21	32.48	0.34	0.40	-6.52	-9.15
BTLG	92.901	23.175	29.85	33.79	0.32	0.37	-5.88	-7.78
KASH	92.869	24.814	29.96	36.99	0.32	0.36	-5.77	-4.07
SRCP	92.856	23.329	30.70	34.85	0.49	0.58	-5.03	-6.67
AZWL	92.732	23.724	29.12	35.35	0.61	0.73	-6.61	-6.02
KLSB	92.731	24.371	31.18	34.59	0.24	0.27	-4.55	-6.57
BRPI	92.685	23.306	31.04	33.38	0.42	0.48	-4.69	-8.10
MAMT	92.49	23.943	31.37	34.35	0.20	0.24	-4.36	-6.89
IGS Stations								
WUHN	114.357	30.532	-12.04	32.31	0.11	0.20		
IRKT	104.316	52.219	-6.69	25.08	0.12	0.10		
NTUS	103.680	1.346	-12.42	13.99	1.13	1.33		
KUNM	102.797	25.03	-19.24	30.41	0.40	0.49		
COCO	96.834	-12.188	52.88	44.68	0.23	0.20		
LHAS	91.104	29.657	16.12	45.12	0.55	0.21		
GUAO	87.177	43.471	5.70	31.79	0.18	0.13		
HYDE	78.551	17.417	34.75	40.94	0.08	0.29		
IISC	77.570	13.021	35.00	42.85	0.08	0.46		
BAN2	77.512	13.034	35.36	43.52	0.14	0.41		
DGAR	72.370	-7.270	33.27	46.70	0.12	0.16		
KIT3	66.885	39.135	5.57	27.60	0.15	0.18		
BAHR	50.608	26.209	30.80	29.83	0.36	0.20		

Table S2. Comparison of Euler poles for Indian plate motion. Modified from Mahesh et al., (2012).

Euler Pole Reference	Latitude °N	Longitude °E	Angular velocity °/My
Mahesh, 2012 India/ITRF2008	51.4±0.07	8.9±0.8	0.539±0.002
This work India/ITRF2008	51.42±0.06	2.10±0.38	0.5146±0.001
SOPAC Website India/ITRF2000	53.1	2.2	0.519±0.019
Banerjee et al. (2008) India/ITRF2000	52.9±0.21	−0.2±3.7	0.499±0.008
Bettinelli et al. (2006) India/ITRF2000	51.4±1.6	−10.9±5.6	0.483±0.01
Socquet et al. (2006) India/ITRF2000	50.9±5.1	−12.1±0.6	0.486±0.001
Sella et al. (2002) India/ITRF1997	53.7	−13.9	0.483
Jade et al. (2007) India/ITRF2000	51.7±0.5	−15.1±1.5	0.469±0.01

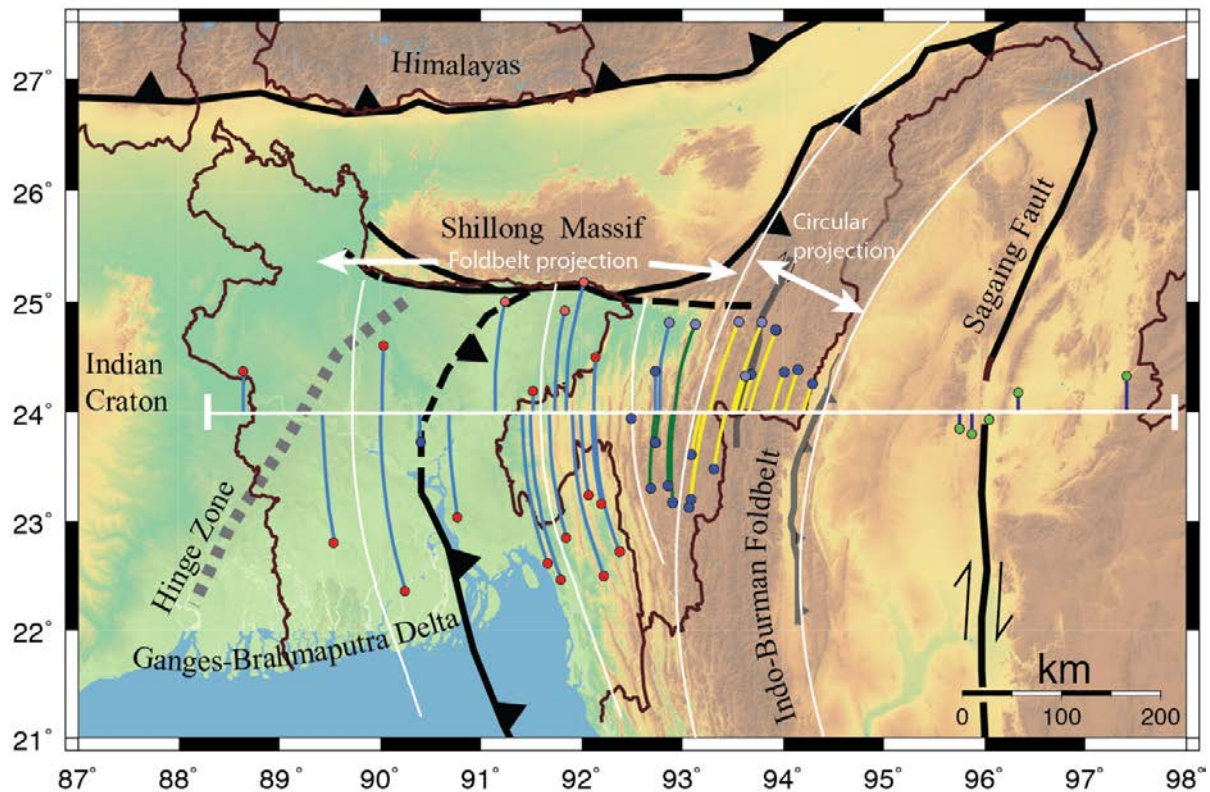


Figure S3. Map illustrating the projection of the GPS velocities onto a line. The Myanmar stations are simply projected north and south. The backstop region of the foldbelt in the higher Indo-Burman Foldbelt are projected using a circle centered at 22.5° N, 99.5° E (yellow lines). The shortening and shear directions are taken as perpendicular and parallel to the circle passing through the site. The distance along the profile is determined by continuing along the circle to the profile at 24°N. The sties in the foldbelt are projected using a surface constructed from the arcs of the fold and the alignment of the segment boundaries between anticlines as the transport direction (blue lines). The white lines show some illustrative paths. For five stations in the transition between the two projections, we averaged the results of the two projections (green lines). Station colors are as in Figure 1, Red- Bangladesh, Blue – India, Green – Myanmar.

Models

For all of the elastic loading models we fixed the shear modulus at 30 GPa, and Poisson's ratio at 0.25, although the results are not sensitive to these values. To find the best fitting megathrust model, we ran sets of models in which we fixed the dip and locking depth, then computed a least-squares fit of the fault position and slip rate. We explored possible dips from 5° - 20° and locking depths from 15 - 50 km. While the location of the surface projection of the fault varied, the horizontal position of the downdip end of the locked zone stayed with ~1 km for all of good fitting models. Its depth traded off with dip, such that shallower dips yielded shallower locking depths. To better constrain the dip, added a misfit from the dip that best fit the seismicity at the end of the locked zone, 0.0005 ($dp - 7.3$). The best fitting model is 25 km depth, 9° dip (Fig. S4).

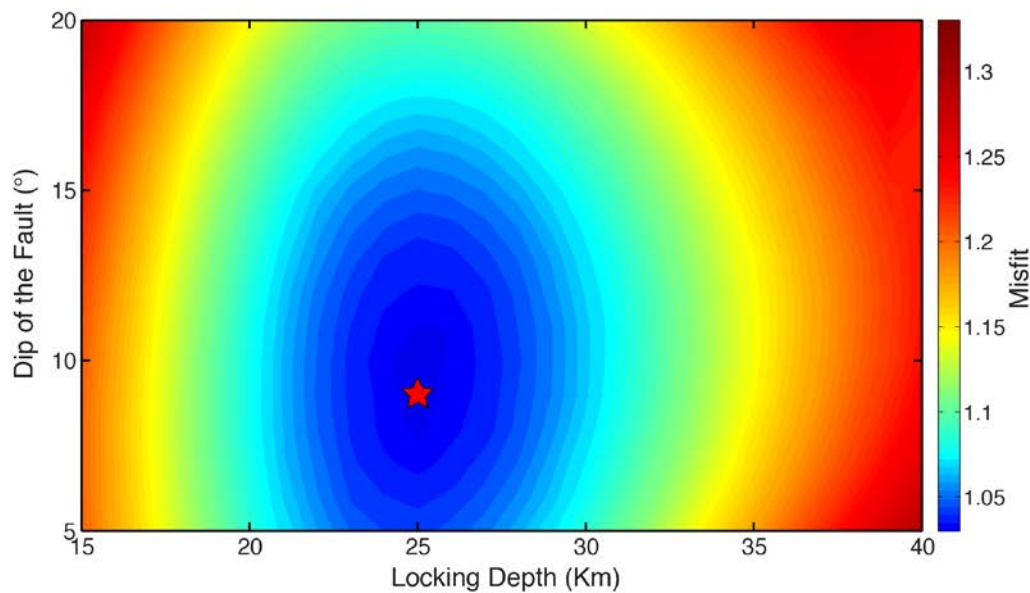


Figure S4. Misfit as a function of dip and depth of the end of the locked zone based on combined fit to interseismic GPS data and slab seismicity. The depth constraint comes primarily from the GPS data and the dip from the seismicity. The red star marks the preferred model, 25 km depth and 9° dip.

As our focus is on establishing convergence, we performed simpler models for the strike-slip faults in order to account for the GPS velocity field. Sagaing Fault was modeled as a locked strike-slip fault using a locking depth of 6.3 km from Maurin et al.⁹ with the dextral slip rate increased slightly to better fit the GPS data.

Since there are limited data to constrain the locking depth for the Churachandpur-Mao Fault, it was modeled as a locked strike slip fault using same locking depth of 6.3 km as for the Sagaing Fault and only the slip rate was fit to GPS data.

The Kabaw Fault has no GPS constraints, and was modeled as 30° E dipping fault with slip rates of thrust and strike slip components to fit the offset between Indian and Myanmar GPS data.